

CHEMICAL REACTIONS

In previous classes, we have learnt about atoms as the smallest particle of any matter. Also, elements as pure form of matter, and the way they interact with each other. Do you think that all these interactions always result into new compounds? No. we have already discussed that these interactions may result in either physical or chemical change. In this chapter, we will study the conditions necessary for any chemical change, types and nature of chemical reactions along with its significance in daily life. Let's explore on the basis of our prior knowledge. Do you think melting of ice is a chemical change? Have you observed fire as a result of burning coal in the presence of oxygen? Can you get back the paper which has been burnt? Why we inhale oxygen gas and exhale

In this Chapter you will learn about:

- Chemical Reactions (Definition & Applications)
- Chemical Equation and Balancing
- Law of Conservation of Mass
- Types of Chemical Reactions (Addition and Decomposition)
- Energy Changes in Chemical Reactions (Exothermic and Endothermic)

All the students will be able to:

- Define chemical reactions and give examples.
- Explain the rearrangement of atoms in chemical reactions.
- Explain the balancing of a chemical reaction.
- Define the law of conservation of mass.
- Identify the nature of a chemical change in various reactions.
- Describe changes in the states of matter in a chemical reaction.
- Explain the types of chemical reactions with examples.
- Explain the energy changes in chemical reactions.
- Describe the importance of exothermic reactions in daily life.

**Fig.5.1: Combustion reaction****Fig.5.2: Rusty Iron**

carbon dioxide during respiration? Which of these interactions result in the formation of new compounds whose properties are different than the original compounds?

CHEMICAL REACTION

- Define chemical reactions and give examples.
- Explain the rearrangement of atoms in chemical reactions.
- Describe changes in the states of matter in a chemical reaction.

Do you know what happens to coal when it is heated in the presence of oxygen? It results in the fire (heat) and evolution of gas like CO_2 . Coal is black coloured solid form of carbon whereas it produces CO_2 which is a colourless gas. This is an example of chemical change where product has different chemical composition and properties than the substance reacted. Also, the change is permanent and cannot be reversed. Hence, we can say that coal and oxygen reacted chemically to produce CO_2 . Such process is called chemical reaction. Few more examples are;

Vinegar + Soda _____ Sodium bicarbonate

Iron nail + water _____ Rusting

Iron + Sulphur _____ Iron sulphide

Hence, we can say that

“A chemical change is called a chemical reaction”

The rearrangement of atoms takes place during chemical reactions. For example, in the above case, an atom of carbon reacts with a molecule of oxygen to form a molecule of carbon dioxide with the evolution of heat. This is shown in the figure 5.3

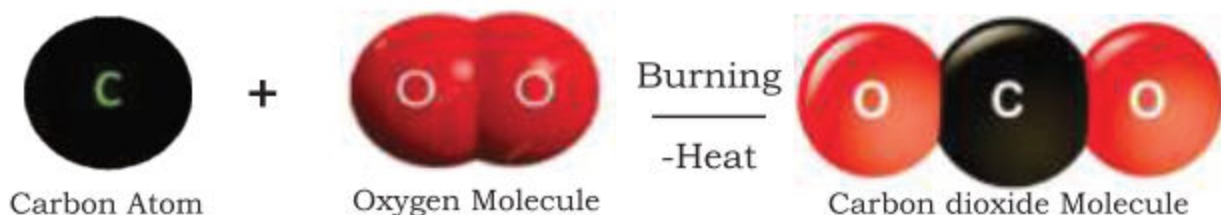


Fig.5.3:

Combustion of methane gas is shown in Fig. 5.4. During this reaction, methane (CH_4) gas reacts with oxygen (O_2) gas to form carbon dioxide (CO_2) and water (H_2O). Rearrangement of atoms take place during this process. Carbon atom of methane (CH_4) gets attached with two oxygen atoms to form carbon dioxide, whereas two H atoms of methane get attached with oxygen atom to form water.

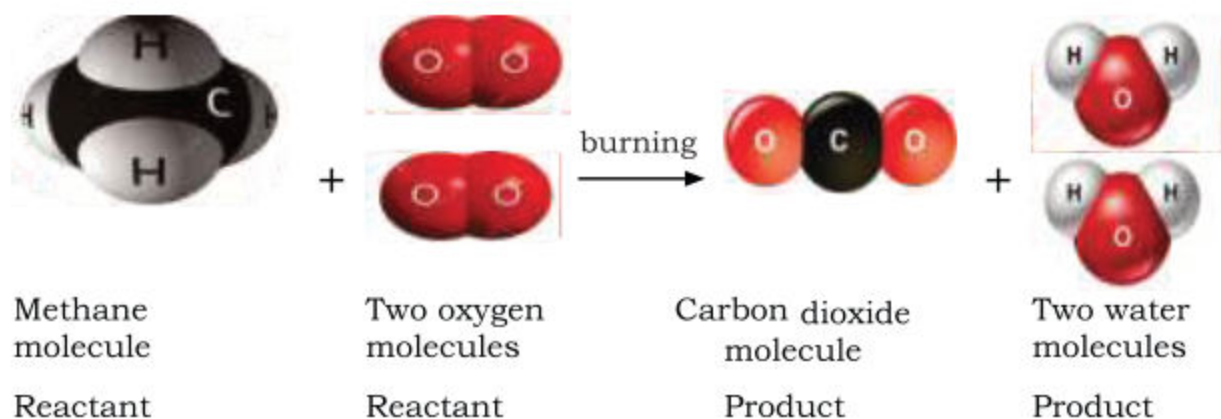


Fig. 5.4

Chemical reactions are usually expressed in the form of chemical equation where the substances take part in the reaction are called reactants and are written on the left side of the arrow. Whereas the substances formed are called products and are written on the right side of the arrow. We can conclude that chemical reaction:

- produces new substance which may differ in the properties than the original substance.
- occurs with absorption or emission of energy mostly in the form of heat
- may be fast, moderate or slow.

When writing chemical equations, chemists often indicate the physical states of reactants and products by using the abbreviations g, l, s and aq in parentheses to denote gas, liquid, solid, and the aqueous (water) environment. States of matter changes during chemical reactions. For example,



Activity 5.1:

Rearrangement of Atoms

By using any of the above two equations, express the rearrangement of atoms in a chemical reaction as shown in figure 5.3

Activity 5.2:

Identification of Chemical Reactions

Identify which of the following are chemical reactions or physical change and justify your answer in the box given below

1. Burning of paper _____
2. Burning of candle _____
3. Baking of Cake _____

Applications of Chemical Reactions

This book, our body or food everything is made of chemical matter. Most of the changes in our surroundings are due to chemical reactions. For example, every time we cook, light a match stick, take a breath, or consume an antacid to get relief from stomach acidity, we carry out chemical reactions. Knowing some chemistry can help us in making day to day decisions that affect our lives. Like mixing of two chemicals at home.

DO YOU KNOW?

Chemical reactions do not only take place in laboratory, but they occur all around us

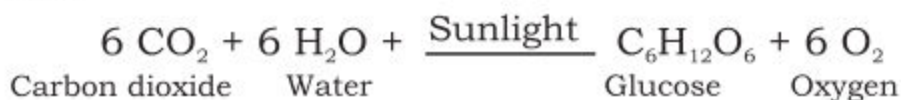
Exploration

Can you identify few changes in your surrounding which can be categorized as chemical reactions?

Chemical reactions can be beneficial or hazardous to our health, environment or society in general. For example,

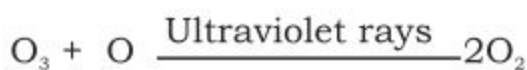
Photosynthesis

Green plants prepare their food by the process of photosynthesis. In this chemical reaction, leaves convert carbon dioxide and water into food in shape of glucose and oxygen. It is one of the most common everyday chemical reactions and also one of the most important since this is how plants produce food for themselves and for the animals who eat these plants. Also, in this reaction oxygen is produced.



Ozone Layer Depletion

Ozone layer is a belt of naturally occurring gas Ozone (O_3) above the surface of earth and serves as a shield to protect our earth from the harmful ultraviolet rays from the Sun. However, this layer is getting destroyed due to the chemical reaction with compound like Chlorofluorocarbons (CFCs) that are released from industries, and aerosols like body sprays, perfumes.



CHEMICAL EQUATIONS AND BALANCING

- Explain the balancing of a chemical reaction.
- Balance simple chemical equations.

As we have already discussed that chemical reactions are represented by chemical equations in which chemical symbols are used to denote different elements and compounds as reactants and products. This is time to discuss as how and why we balance chemical equations. Take an example of following equation



In this equation, it is represented that hydrogen gas and oxygen gas when react, form water. However, this expression is not complete, because there are twice as many oxygen atoms on the left side of arrow as on the right side. According to law of conservation of mass (which we will discuss later), there must be the same number of each type of atom on both sides of the arrow; that is, we must have as many atoms after the reaction ends as we did before it started. So, we can balance this expression by placing an appropriate coefficient (2 in this case) in front of H_2 and H_2O .



This is to note that we balance a chemical equation by changing the coefficients, never the subscripts. Changing the subscript would change the identity of the compound. There is no one formula that we can use for balancing the equations, rather we do it with trial and error method. We can follow the steps given below while balancing any chemical reaction.

- Write down the formula of reactants to the left of the arrow, and the formulas of the products to the right of the arrow.
- Once you have correct formulas of the reactants and products, begin balancing the equation by trying suitable coefficients that will make the number of atoms of each element the same on both sides of the equation.

- First look for elements that appear only once on each side of the equation and with equal numbers of atoms on each side. The compounds containing these elements must have the same coefficients. Next, look for elements that appear only once on each side of the equation but in unequal numbers of atoms. Balance these elements. Finally, balance elements that appear in two or more compounds on the same side of the equation.
- Check your equation if it has the same total number of each type of atom on both sides of the equation arrow.

Consider the reaction of iron with hydrochloric acid to produce iron chloride and hydrogen gas. First write reactants on the left and products on the right side.



We can see that all three elements (Fe, Cl and H) appear only once on each side of the equation, but only Fe appears in equal numbers of atoms on both sides. Therefore, Fe and FeCl₂ must have the same coefficient which is 1. The next step is to balance either the number of Cl atoms or H atoms on both sides of equation (you can choose any). To balance Cl, we place the coefficient 2 in front of HCl.



We see for final check that the number of atoms of each element in both the reactants and products are same, which means that the equation is balanced.

Elements	Reactants	Products
Fe	1	1
H	2	2
Cl	2	2

Activity 5.3:

Balancing the Equations

Balance the following equations by rewriting these with appropriate coefficients in the space given below.



Exploration

Do you know why balanced chemical equations are important to scientists?

LAW OF CONSERVATION OF MASS

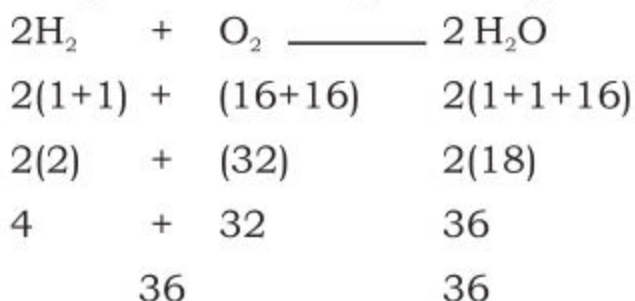
➤ Define the law of conservation of mass.

In 1789, a French chemist, Antoine Lavoisier, who is known as the father of modern chemistry, put forward the law called “law of conservation of mass”. According to him

In a chemical reaction, the mass is neither created nor destroyed but it changes from one form to another form or in other words, mass of reactants and products remains constant.

For example, consider the formation of water molecule. Two hydrogen molecules combine with one oxygen molecule to form two water molecules. Calculating the mass of both sides of

reactions (reactants and products), we will find these exactly same.



DO YOU KNOW?

Atomic mass of H=1, and O=16. Use these to verify the law of conservation of mass in balanced chemical equation

Activity 5.4:

Proving Law of Conservation of Mass through Calculations

Calculate mass of reactants and products in the given reaction. Important to note that your equation needs to be balanced.

So first step is to check whether the given chemical equation is balanced or not.



Atomic mass C = 12 , H = 1 O = 16 , Na = 23

TYPES OF CHEMICAL REACTIONS

- Identify the nature of chemical change in various reactions.
- Explain the types of chemical reactions with examples

There are different types of chemical reactions based on what happens when reactants change into products. Here we will discuss two basic types of chemical reactions i.e. addition reactions and decomposition reactions.

Addition Reactions

Those chemical reactions in which two or more reactants combine to form a product are called addition reactions. These are also called as combination reactions or synthesis reactions because in these reactions two or more substances combine and synthesize one substance. For example, the reaction of sodium and chlorine

to form sodium chloride,

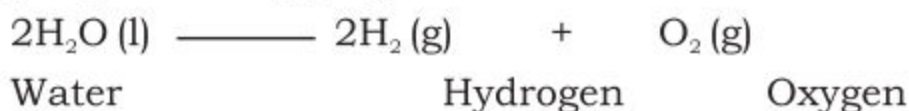


and the burning of coal (carbon) to give carbon dioxide,

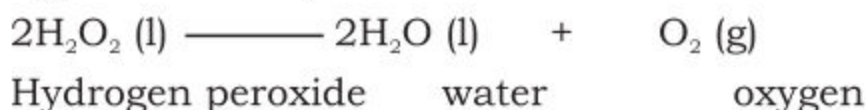


Decomposition Reactions

Decomposition reactions are the opposite of addition reactions. In decomposition reactions, a single compound breaks down into two or more simpler substances. For example, the decomposition of water into hydrogen and oxygen gases.



Another example is decomposition of hydrogen peroxide to form oxygen gas and water,



Decomposition reactions happen spontaneously in unstable compounds, however, require external conditions like heat, catalyst or electric current to decompose the stable compounds.

Activity # 5.5: Addition or Decomposition Reactions

Identify which of the following reactions are addition reactions or decomposition reactions. Write your answer in the column next to each equation

Chemical Reactions	Type of Reactions
$2\text{Mg} + \text{O}_2 \longrightarrow 2\text{MgO}$	
$2\text{KClO}_3 \longrightarrow 2\text{KCl} + 3\text{O}_2$	
$\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$	
$\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$	
$\text{CaO} + \text{CO}_2 \longrightarrow \text{CaCO}_3$	
$\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$	

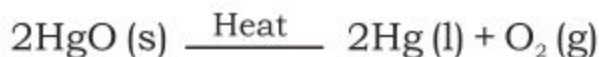
ENERGY CHANGES IN CHEMICAL REACTIONS (EXOTHERMIC AND ENDOTHERMIC)

- Explain the energy changes in chemical reactions.
- Describe the importance of exothermic reactions.

Almost all chemical reactions absorb or release energy. Heat is the form of energy most commonly absorbed or released in chemical reactions. The study of heat changes in chemical reactions is called “thermochemistry”. Based on the changes in the heat content in reactants and products, we have categorized chemical reactions into two types.

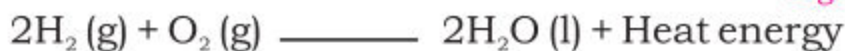
Endothermic Reactions

Endothermic reactions are those reactions in which heat energy is absorbed or added to the system. “Endo” means inside whereas “therm” means heat. One common example is the decomposition of mercury oxide (HgO) at high temperature.



Exothermic Reactions

Exothermic reactions are those reactions which release or give off heat. “Exo” means outside whereas “therm” means heat. One common example is combustion of hydrogen gas in oxygen which releases considerable quantities of heat energy



Exothermic reactions are very common and are significantly important in everyday life. When we take food, it gives out energy for the cell to function and to make proteins and new cell and hence our body grows. Without the exothermic reaction, every living cell would die due to the unavailability of energy.



Fig.5.5: Exothermic Reaction

Similarly, exothermic reactions can be used for everyday purposes. For example, thermite reaction which involves converting iron oxide to iron by reacting it with aluminum oxide. This reaction is most commonly used to repair cracks in railway lines.

Activity 5.6:

Exothermic & Endothermic Reaction

Materials needed: clear glass or cup, vinegar, baking soda, thermometer, water, common salt.

Steps:

1. Fill half the cup with vinegar (liquid).
2. Put thermometer to measure the temperature of liquid. Record it below.
3. Add a tablespoon of baking soda. Stir and wait for a minute and then record the final temperature of a substance.
4. Record the result of experiment 1.
5. Wash the apparatus with water and repeat the same experiment with water (as liquid) and common salt.
6. Record your result of experiment 2.

Experiment 1: Vinegar and Baking soda

Initial Temperature: _____ °C

Final Temperature: _____ °C

Difference: Final temperature – initial temperature = _____ °C

Experiment 2: Water and Common salt

Initial Temperature: _____ °C

Final Temperature: _____ °C

Difference: Final temperature – initial temperature = _____ °C

Question:

Comparing the initial and final temperatures, what do you think is the type of both reactions? Why?

DO YOU KNOW?

Combustion, neutralization and respiration are few examples of exothermic reactions. Can we imagine life on this earth without these reactions?

Exploration: Why do you require more food for working?

our body is an exothermic engine? It's how we maintain our body temperature at 98.6°F when the surrounding environment is only 65°F. Can you think where this energy in our body comes from? Why do we feel hungry more in cold season?

SUMMARY

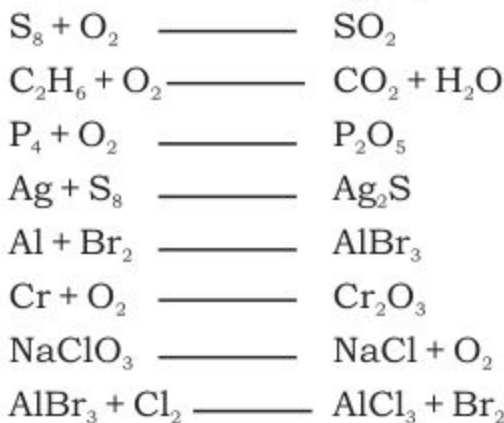
- In a physical change the shape, size, appearance or state of a substance may alter, temporarily. It is usually reversible and no new substance is formed.
- In a chemical change, new substances are formed and energy changes are involved. There is a permanent change takes place in chemical reaction.
- Chemical reactions take place all around us, few unstable substances react spontaneously whereas others require heat, light or catalyst to initiate the reactions.
- The chemical formula of a substance is the symbolic representation of the actual number of atoms present in one molecule of that substance.
- A complete chemical equation represents the reactants, products and their physical states symbolically.
- Equations must always be balanced. Equations are balanced by the trial and error method.
- Total mass of the reactants and products will remain conserved during a chemical reaction, this statement is called law of conservation of mass.
- Chemical reactions that proceed with evolution of heat energy, are called exothermic reactions.
- Chemical reactions that proceed with the absorption of heat energy are called endothermic reactions.
- Addition or combination or synthesis is a reaction in which a new single compound is formed by the union of two or more substances.
- In a decomposition reaction, a compound breaks down into two or more simple substances by the application of heat or electricity.

EXERCISE

1. Define Chemical reactions? Also, write its examples in daily life?
2. Write four conditions which show the occurrence of chemical reaction.
3. Which ones of the following is physical (P) or chemical (C) change.

- | | |
|---|-------|
| a. Rusting of locks | _____ |
| b. Melting of ice | _____ |
| c. Change in taste of milk | _____ |
| d. Digesting your food | _____ |
| e. Change in meat's smell after it is out in the sun for some time. | _____ |
| f. Carving on a piece of wood | _____ |
| g. Mixing of sugar in water | _____ |
| h. Mashing potatoes | _____ |
| i. Lighting a match | _____ |

4. Balance the following equations.



5. Choose the correct answer.

- a. What is an example of a synthesis reaction?
 - A. CO_2 breaking down to $C + O_2$
 - B. $Zn + 2HCl$ _____ $ZnCl_2 + H_2$
 - C. Hydrogen and Oxygen combining to form water.

- b. Which is the BEST indication that a chemical reaction has taken place?
- A. Blue powder turns water blue.
 - B. A white substance dissolving in water.
 - C. Metal bubbling when placed in an unknown liquid.
- c. What are the products in the reaction: $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- A. Carbon Dioxide and Water.
 - B. Carbon Monoxide and Hydroxide.
 - C. Salt and Vinegar.
- d. Which of the following is an example of a chemical reaction?
- A. salt crystals forming as sea water evaporates.
 - B. paper turning soft when wet.
 - C. leaves changing colours in the fall.
6. Compare exothermic and endothermic reactions. Write down similarities and differences?
7. Write at least two examples of addition and decomposition reactions.
8. Describe the importance of exothermic reactions.
9. Write down the chemical reactions of the following in the form of balanced chemical equations.
- a. Carbon dioxide (CO_2) and water (H_2O)
 - b. Carbon monoxide (CO) and oxygen (O_2)
 - c. Zinc (Zn) and hydrochloric acid (HCl)
 - d. Hydrochloric acid (HCl) and sodium hydroxide (NaOH)
 - e. Decomposition of potassium chlorate (KClO_3)